

# Threading

## Extending the Thread class

Threads can be created by extending the `Thread` class.

## Creating Threads Using Lambda Expressions

Threads can be created using lambda expressions.

## Purpose of Threads

Threads are used in Java to improve performance for processes that can be split into code that can run concurrently.

## Java Thread Lifecycle

A Java thread can, throughout its lifecycle, be in New, Running/Active, Blocked, Waiting, or Terminated/Joined states.

## Thread Communication in Java

Threads can communicate with each other in Java by polling the state of a shared resource.

## Java Thread Lifecycle Methods

A programmer can control Java thread lifecycle states using the following methods:

- `.start()`
- `.sleep()`
- `.join()`
- `.isAlive()`

## Threads Communication Methods In Java

Threads can communicate with each other in Java using the `.wait()`, `.notify()` and `.notifyAll()` methods.

## Synchronized in Java

In Java, the `synchronized` keyword is used to ensure that methods in two different threads do not access a shared resource at the same time.

## Thread

A thread is a part of a program that follows a path of execution independently. Multiple threads can run at the same time (concurrently).

## Race Conditions in Java Threads

A race condition occurs when a multi-threaded program produces unexpected behavior due to the timing or interleaving of the execution of threads.

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